

- Motor control is kind of localized to one area
 - Primary motor cortex (M1, precentral gyrus)
 - The homoculus indicates where the hand is . Not sure if we have that much localization with EEG
 - All the pre-planning and intention is distributed throughout the brain, not just there
- Contralateral organization is key
 - Left side of the brain = right hand
 - Right side of brain = left hand
 - Keep the waves intentionally broad (not localized to one area) so that the pre-planning/intention signals are picked up as well
 - Create a classifier/detector for an ERD, and apply it to both sides, then see which side has a significantly higher probability?
- Concerns with execution dataset
 - Includes artifacts from EMG (electric signals from muscles)
 - Some poor readings due to having to movement during it as well
- Concerns with imagined dataset
 - Less strong signal
 - Some people have struggles with imaging (up to 30%)
- Finding the correlates between the imagined dataset and execution dataset gets you the relevant signal!
- Datasets and subjects
 - There's a lot of subject differences. I'm not sure if the best approach is to control by subject or to do the opposite (mix everything)

Todos

- Neuroscience research
 - How contralateral is contralateral?
 - Neuroscience: The contralateral in the brain is mostly true, but some factors that complicate it: left-handedness, movement force, complexity of task (which is constant)
 - If the user is imagining, it also becomes more complicated and involves a wider, more bilateral network in the brain
 - Bilateral effects
 - If contralateral side increases, frequently the ipsilateral side decreases as well
 - **“push–pull contrast: ERD on the active side *and* frequently ERS on the other”**
 - Due to EEG, there is a lot of overlap/go-reporting between the two sides
 - Sensorimotor rhythms
 - Higher rhythm when inactive

- Oscillate in a rhythmic way
 - mu (~8–13 Hz) and beta (~13–30 Hz) bands.
- When motor control is active → event-related desynchronization
 - The voltage read goes down?
- My thoughts: we have some quantification of history before the event, not just the epoch, so we can quantify whether it was inactive before and active now
- **How do artifacts show up in the data and how would the types of artifacts differ if we looked at the arms/legs. If they are similar, we can explore how to remove the artifacts using the union of arms/legs and hand trials**
- Methods research
 - What preprocessing methods are good for EEG and in what cases
 - **How to find neural correlates of activity**
- Data exploration
 - Look at the EEG data, understand how many electrodes I have
 - Select which ones to keep
 - Start off with only C3 and C4, the electrodes that sit on top of lateralized region relating to muscular control
 - Then for more complicated steps, add some sort of spectral method so that the signal is more clear/localized to C3 and C4, and leakage from neighbors is minimized
 - Laplacian or Reimann method
 - Look for visual differences around the markers
 - Compare 2 subjects at rest and in execution
 - Compare 2 subjects in execution and imagination
 - **Conclusion:** From visual comparison, I can't seem to notice
 - **From this identify**
 - The rough time window for epochs
 - **Investigate intra-subject variance**
 - Compare the same epoch across many subjects, and see how it differs.
 - Findings
 - Peoples baselines for the mu/beta rhythm is different
 - Solution: fit a personalized baseline per person. The concern with this is that for a new testing example, we only have the one sample to base the entire baseline off of and so it may be very biased. I guess in some sense it wouldn't overfit to the data/thats what we want
 - Claude suggested a cheap approach, and a more complicated approach using CPS or covariance networks
 - "BCI illiteracy" impacts 15-30% of people where they can't imagine what they're doing

- fatigue, attention, electrode drying can impact rhythm strength
- Comparisons for later
 - Strong localization to brain regions with EEG vs general sensorimotor rhythms (involve more electrodes)

Things to investigate

- Neural correlates
 - Get rid of the brain wave activity associated with the physical movement, independent of location (difference between legs and arms), or whatever is the same between left hand and right hand
 - The only viable idea: compare the left hand right hand mu waves with the mu waves in the electrode closest to foot movement (according to homunculus) and the one closest to hand movement
 - to assess whether the signals have somatotopic organization, and thus that there is a lower chance of artifacts due to other brain activity
 - Finding neural correlates of the hand clenching behavior is not needed because it doesn't show what lateralized EMG looks like. I guess what Claude is saying is that it's a completely different task, neurologically, than just doing one side.
- Artifacts
 - EMG traces because clenching muscles requires neural firing and that can be picked up
 - Has higher amplitude usually

Experiment Design Notes

- 4.1s window between triggers to contract muscle

The make up of an ERD

- Decrease in the power of a particular brainwave
 - The motivation is that when the neurons are at rest they fire synchronously, but when become task specific, they become independent, not synchronous, and thus the summation of them decreases and they're less adherent with the wave
 - → we should feed in ERD for mu and beta bands

Data

- Start off with only C3 and C4, the electrodes that sit on top of lateralized region relating to muscular control

Feature Set

- We have some quantification of history before the event, not just the epoch, so we can quantify whether it was inactive before and active now
 - Per side, to detect which side is more likely to be the case now
- Compare the two regions (C3 and C4): “push–pull contrast: ERD on the active side and frequently ERS on the other”
 - ERS on the contralateral side because of resetting to baseline
 - ERS on the ipsilateral side because of push-pull contrast
 - ERD on contralateral side because neurons are becoming focused

Preprocessing steps

- For more complicated steps, add some sort of spectral method so that the signal is more clear/localized to C3 and C4, and leakage from neighbors is minimized
 - Laplacian or Reimann method
- Use general trends/knowledge about EMG (that its higher voltage than the other signals) to filter it out
 - I wanted to do something personalized based off of the other trials of the model (like seeing the common brain activity between closing 2 fists and the one by one trial)

Prediction/output clarifications

- For a session with already marked hand clenches, mark whether they are right or left hand.
 - The timestamps are already there

Edge case/artifacts/thoughts

- When there are multiple of one hand movement in a row, what is the recovery time?
 - Rebound time is up to 5 seconds after the trigger
 - ~0 to ~3–4 s post-cue (the ERD period) plus optionally ~4–5 s to catch the rebound
- Task complexity influence on contralateralness is not as much of a concern since its being held constant
- Left handedness vs right handedness

Process

1. Start with just execution training set
2. Then expand to imagination as well
3. Experiment with whether to include the baseline or not

Potential design

- Train one model/algorithm to compute someone's baseline from their EFG sample

For evaluation

- Check per class performance to see if its discriminating against left handed people (who are less represented) because using our non dominant hand is more integrated
 - Is there a way to predict if someone is left handed or right handed, and drop the people who are assumed to be right handed
 - Don't want the model to learn a right-skew bias? Can normalization help with this?

My general thoughts

- My working question is how to translate all these neuroscientific findings into code
- Can I compute some baseline about the user from the whole sample to personalize it to them
- I used Claude a lot, but ignored a lot of what it was saying
- How to process/guide the LLM to understand the tradeoff

Additional things to add

. **Attention and the cue itself are confounds**

- Mu/alpha is suppressed by **attention and effort**, not only by movement. And each trial is **cued**, so the cue evokes visual/attentional responses (visual evoked potential, a preparatory slow potential).

- If left vs. right trials differ in attention or in cue properties, some "decoding" could be attentional rather than motor. Usually left/right are balanced, so this is second-order — but it's why the **somatotopy and eyes-closed controls** matter: they show the signal is motor and central, not attentional/posterior.

5. Non-stationarity within a run: habituation and fatigue

- ERD **attenuates with repetition/practice and with fatigue** over a run. So the signal isn't identical trial 1 vs. trial 30.
- Adds to the session/subject drift already discussed — a reason trial order can matter and a reason your baseline should track local rest rather than a single global estimate.

Artifacts in our waves

- Mu/alpha is also impacted by attention + effort
 - Can we identify what signal is noise from this and what is real using the other controls?
- **Need to have historical factors and some sort of model for fatigue across a trial, residue from previous trials, and**
 - "session/subject drift"

Other questions

- Does our current model include beta and mu or just mu?

Note on different waves

- Mu = indicates resting/disengaged state of sensorimotor cortex.
 - At rest = present
 - Engaged = lower magnitude
 - What engages it
 - **Actual movement**
 - **Motor imagery**
 - Somatosensory stimulation
 - **Somatosensory/motor attention and anticipation**
 - Anticipating an action and thus attending to it before even doing the action (imagination)
 - Design decision: have the time window include the -2 seconds before, to see if any ERD is found.
 - Baseline needs to be well established in order to accomplish this (issue is that the rest periods also include some residue), need to look at the mechanics of the experiment to see how the experiment reduced attention artifacts (like how did they cue)
- Beta = indicates whether change in the motor cortex is incoming
 - High = motor cortex can stay steady, no change is looming

Mu vs. beta — the useful contrast

	Mu (~8-13 Hz)	Beta (~13-30 Hz)
Leans	somatosensory	motor output
Indexes	cortex engaged / idling	motor "status quo" maintenance
Signature	sustained ERD during action	ERD during, strong rebound after
Also driven by	somatosensory attention, observation, tactile input	top-down control, effort, cognitive set, tonic drive
Attention effect	anticipatory/spatial attention to limb suppresses it	cognitive load / control / preparation modulate it

How attention and effort ride on each — and why it's a confound

- **Both desynchronize with engagement, but for partly different reasons:** mu drops with **sensory/limb attention and anticipation**; beta drops with **preparation, effort, and top-down control**. Neither drop is *proof* of movement.
- **Effort asymmetry is the concrete risk:** if one hand's imagery/movement is harder (handedness), it draws more attention (extra mu ERD) *and* more control/effort (extra beta ERD). Both bands then show a lateralized difference that's **partly effort, not motor engagement** — and it aligns with your class label, so the classifier can't separate them.
- **The rebound helps you here:** the **post-movement beta rebound is fairly motor-specific** — a pure attention/effort account doesn't naturally predict a sharp beta *overshoot at movement offset*. So a lateralized beta rebound is stronger evidence of genuine motor processing than a mere ERD during the trial.

Implication for your features

- **Use both bands, but know they carry different information:** mu gives you sustained sensorimotor engagement; beta adds motor-specific structure including the rebound. A filter bank that keeps mu and beta separately (rather than one 8-30 lump) lets you see which band actually drives your left/right decoding per subject.
- **The rebound is a feature *and* a control:** including a window that captures the beta rebound can add motor-specific signal and simultaneously argues against a pure-attention confound.
- **For the correlate/artifact argument:** "the effect is central, somatotopic, in mu *and* beta, and shows a movement-locked beta rebound" is a much harder claim to dismiss as attention/effort than "8-13 Hz power dropped."